

CS 4032 – Web Programming



Bootstrap

What is Responsive Web Design?

- Responsive web design is about creating web sites which automatically adjust themselves to look good on all devices, from small phones to large desktops.
- Bootstrap is the most popular HTML, CSS, and JavaScript framework for developing responsive, mobile-first web sites.
- Bootstrap is completely free to download and use!

Bootstrap

- Introduction, overview, history
- How to get bootstrap
- Bootstrap grid system
- Bootstrap Typography
- Bootstrap containers
- Bootstrap tables
- Bootstrap Images
- Bootstrap Jumbotron
- Bootstrap Badges
- Bootstrap List, Dropdowns
- Bootstrap Buttons
- Bootstrap Inputs and Forms
- Bootstrap Nav, NavBars, Tabs
- Bootstrap Cards
- Bootstrap Collapse, Accordion
- Bootstrap Pagination
- Bootstrap popovers, Alerts
- Bootstrap Progress Bar, Spinners
- Bootstrap Modal
- Bootstrap Tooltip, Toast
- Bootstrap Media
- Bootstrap Carousel

Bootstrap

- Bootstrap is one of the most popular Front End Framework with modern responsive web design(RWD). HTML, CSS, and JavaScript form the core base on which this framework is based.
- Bootstrap has predefined classes where you can directly apply on the HTML tag without writing lengthy code by including Bootstrap libraries in the project.

Bootstrap History

- Bootstrap was developed by Mark Otto and Jacob Thornton at Twitter, and released as an open source product in August 2011 on GitHub.
- Advantages of Bootstrap:
 - **Easy to use:** Anybody with just basic knowledge of HTML and CSS can start using Bootstrap
 - **Responsive features:** Bootstrap's responsive CSS adjusts to phones, tablets, and desktops
 - **Mobile-first approach:** In Bootstrap 3, mobile-first styles are part of the core framework
 - **Browser compatibility:** Bootstrap is compatible with all modern browsers (Chrome, Firefox, Internet Explorer, Safari, and Opera)

Where to Get Bootstrap?

- There are two ways to start using Bootstrap on your own web site.
 - Download Bootstrap from getbootstrap.com
 - If you want to download and host Bootstrap yourself, go to getbootstrap.com, and follow the instructions there.
 - Include Bootstrap from a CDN
 - If you don't want to download and host Bootstrap yourself, you can include it from a CDN (Content Delivery Network).
 - MaxCDN provides CDN support for Bootstrap's CSS and JavaScript. You must also include jQuery.

Bootstrap CDN

- You must include the following Bootstrap's CSS, JavaScript, and jQuery from MaxCDN into your web page.

```
<!-- Latest compiled and minified Bootstrap CSS -->
```

```
<link rel="stylesheet" href="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/css/bootstrap.min.css">
```

```
<!-- Latest compiled Bootstrap JavaScript -->
```

```
<script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.3.7/js/bootstrap.min.js"></script>
```

```
<!-- latest jQuery library -->
```

```
<script src="https://code.jquery.com/jquery-latest.js"></script>
```

- Advantage of using the Bootstrap CDN:
 - Many users already have downloaded Bootstrap from MaxCDN when visiting another site. As a result, it will be loaded from cache when they visit your site, which leads to faster loading time. Also, most CDN's will make sure that once a user requests a file from it, it will be served from the server closest to them, which also leads to faster loading time.

Bootstrap responsiveness

- Bootstrap is mobile-first
 - Bootstrap 4 is designed to be responsive to mobile devices. Mobile-first styles are part of the core framework.
 - To ensure proper rendering and touch zooming, add the following <meta> tag inside the <head> element:

<meta name="viewport" content="width=device-width, initial-scale=1">

- The width=device-width part sets the width of the page to follow the screen-width of the device (which will vary depending on the device).
- The initial-scale=1 part sets the initial zoom level when the page is first loaded by the browser.

Bootstrap Containers

- Containers
 - Bootstrap also requires a containing element to wrap site contents.
 - There are two container classes to choose from:
 - The `.container class` provides a responsive **fixed width container** that is responsive for different devices
 - The `.container-fluid` class provides a **full width container**, spanning the entire width of the viewport
 - **Note:** Containers are not nestable (you cannot put a container inside another container).



Bootstrap Grids

- Bootstrap's grid system allows up to 12 columns across the page.
- If you do not want to use all 12 columns individually, you can group the columns together to create wider columns:

```
<div class="col-md-12">Span 12 columns</div>
```

```
<div class="col-md-6">Span 6</div><div class="col-md-6">Span 6</div>
```

```
<div class="col-md-4">Span 4</div><div class="col-md-8">Span 8</div>
```

```
<div class="col-md-4">Span 4</div><div class="col-md-4">Span 4</div> <div class="col-md-4">Span 4</div>
```

- Bootstrap's grid system is responsive, and the columns will re-arrange automatically depending on the screen size.

Grid Classes

- The Bootstrap grid system has four classes:
 - xs (for phones)
 - sm (for tablets)
 - md (for desktops)
 - lg (for larger desktops)
- The classes above can be combined to create more dynamic and flexible layouts.

Basic Structure of a Bootstrap Grid

```
<div class="row">
  <div class="col-*-*"></div>
</div>
<div class="row">
  <div class="col-*-*"></div>
  <div class="col-*-*"></div>
  <div class="col-*-*"></div>
</div>
<div class="row">
  ...
</div>
```

- First; create a row (<div class="row">). Then, add the desired number of columns (tags with appropriate .col-*-* classes). Note that numbers in .col-*-* should always add up to 12 for each row.

Bootstrap Typography

Bootstrap 4 Default Settings

- Bootstrap 4 uses a default font-size of 16px, and its line-height is 1.5.
- The default font-family is "Helvetica Neue", Helvetica, Arial, sans-serif.
- In addition, all `<p>` elements have margin-top: 0 and margin-bottom: 1rem (16px by default).

Bootstrap Typography

Bootstrap 5 Default Settings

- Bootstrap 4 uses a default font-size of 16px
- The default font-family is:
 - // Windows
 - "Segoe UI",
 - // Android
 - Roboto,
 - // Basic web fallback
 - "Helvetica Neue", Arial,
 - // Linux
 - "Noto Sans",
 - "Liberation Sans",
 - // Sans serif fallback
 - sans-serif,
- In addition, all <p> elements have margin-top: 0 and margin-bottom: 1rem (16px by default).
- <https://getbootstrap.com/docs/5.0/content/typography/>

Bootstrap Typography

Class	Description
<code>.font-weight-bold</code>	Bold text
<code>.font-weight-bolder</code>	Bolder text
<code>.font-italic</code>	Italic text
<code>.font-weight-light</code>	Light weight text
<code>.font-weight-lighter</code>	Lighter weight text
<code>.font-weight-normal</code>	Normal text
<code>.lead</code>	Makes a paragraph stand out
<code>.small</code>	Indicates smaller text (set to 80% of the size of the parent)
<code>.text-left</code>	Indicates left-aligned text
<code>.text-*-left</code>	Indicates left-aligned text on small, medium, large or xlarge screens
<code>.text-break</code>	Prevents long text from breaking layout
<code>.text-center</code>	Indicates center-aligned text
<code>.text-*-center</code>	Indicates center-aligned text on small, medium, large or xlarge screens
<code>.text-decoration-none</code>	Removes the underline from a link
<code>.text-right</code>	Indicates right-aligned text
<code>.text-*-right</code>	Indicates right-aligned text on small, medium, large or xlarge screens
<code>.text-justify</code>	Indicates justified text
<code>.text-monospace</code>	Monospaced text

Bootstrap Text Colors

- Bootstrap 4 has some **contextual classes** that can be used to provide "meaning through colors".
- The classes for text colors are:
 - **.text-muted, .text-primary, .text-success, .text-info, .text-warning, .text-danger, .text-secondary, .text-white, .text-dark, .text-body (default body color/often black) and .text-light.**
- The classes for background colors are:
 - **.bg-primary, .bg-success, .bg-info, .bg-warning, .bg-danger, .bg-secondary, .bg-dark and .bg-light.**

This text is important.

This text indicates success.

This text represents some information.

This text represents a warning.

This text represents danger.

Secondary background color.

Dark grey background color.

Light grey background color.

Bootstrap Tables

- A **basic** Bootstrap table has a light padding and only horizontal dividers.
 - The **.table class** adds basic styling to a table:
- **Striped Rows**
 - The **.table-striped** class adds zebra-stripes to a table:
- **Bordered Table**
 - The **.table-bordered class** adds borders on all sides of the table and cells:
- **Hover Rows**
 - The **.table-hover class** enables a hover state on table rows:
- **Responsive Tables**
 - The **.table-responsive class** creates a responsive table. The table will then scroll horizontally on small devices (under 768px). When viewing on anything larger than 768px wide, there is no difference:

Bootstrap Images

- Rounded Corners
 - The **.img-rounded class** adds rounded corners to an image (IE8 does not support rounded corners):
- Circle
 - The **.img-circle class** shapes the image to a circle (IE8 does not support rounded corners):
- Thumbnail
 - The **.img-thumbnail** class shapes the image to a thumbnail:
- Responsive Images
 - Images comes in all sizes. So do screens. Responsive images automatically adjust to fit the size of the screen.
 - Create responsive images by adding an .img-responsive class to the tag. The image will then scale nicely to the parent element.
 - The **.img-responsive class** applies display: block; and max-width: 100%; and height: auto; to the image:

Bootstrap Jumbotron

- provides an excellent way to **showcase the key content /information** on a web page.
- A jumbotron indicates **a big grey box** for calling extra attention to some special content or information.
- **Tip:** Inside a jumbotron you can put nearly any valid HTML, including other Bootstrap elements/classes.
- ```
<div class="jumbotron">
 <h1>Bootstrap Tutorial</h1>
 <p>Bootstrap is the most popular HTML, CSS...</p>
</div>
```

## Bootstrap Tutorial

Bootstrap is the most popular HTML, CSS, and JS framework for developing responsive, mobile-first projects on the web.

# Bootstrap Jumbotron

- Jumbotrons are no longer supported in Bootstrap 5. However, you can use a `<div>` element and add special helper classes together with a color class to achieve the same effect:

```
<div class="mt-4 p-5 bg-primary text-white rounded">
```

```
 <h1>Jumbotron Example</h1>
```

```
 <p>t labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat..</p>
```

```
</div>
```

# Bootstrap-Badges

- Badges are generally used to indicate some valuable information on the web pages such as important heading, warning messages, notification counter.
- Use the **.badge class** together with a contextual class (like **.badge-secondary**) within `<span>` elements to create rectangular badges.
- E.g badge-primary, badge-secondary, badge-success, badge-danger etc.
- `<button type="button" class="btn btn-primary">  
 Messages </button>`
- [https://www.w3schools.com/bootstrap4/tryit.asp?filename=trybs\\_badges\\_button&stacked=h](https://www.w3schools.com/bootstrap4/tryit.asp?filename=trybs_badges_button&stacked=h)

# Bootstrap Lists and List Groups

- To create a basic list group,
  1. use an `<ul>` element with **class `.list-group`**,
  2. `<li>` elements with **class `.list-group-item`**
- ```
<ul class="list-group">  
  <li class="list-group-item">Picture</li>  
  <li class="list-group-item">Documents</li>  
  <li class="list-group-item">Music</li>  
  <li class="list-group-item">Videos</li>  
</ul>
```

Pictures
Documents
Music
Videos

- ```
<div class="container">
 <h2>Horizontal List Groups</h2>
 <p>The .list-group-horizontal class displays the list items horizontally instead of vertically:</p>
```

- ```
<ul class="list-group list-group-horizontal">  
  <li class="list-group-item">First item</li>  
  <li class="list-group-item">Second item</li>  
  <li class="list-group-item">Third item</li>  
  <li class="list-group-item">Fourth item</li>  
</ul>  
</div>
```

Horizontal List Groups

The `.list-group-horizontal` class displays the list items horizontally instead of vertically:

First item	Second item	Third item	Fourth item
------------	-------------	------------	-------------

Bootstrap-List Groups

- Use the **.active class** to highlight the current item.
- The **.list-group-horizontal class** displays the list items horizontally instead of vertically.
- Contextual classes can be used to color list items
- The classes for coloring list-items are: **.list-group-item-success**, **list-group-item-secondary**, **list-group-item-info**, **list-group-item-warning**, **.list-group-item-danger**, **.list-group-item-primary**, **list-group-item-dark** and **list-group-item-light**.
- List groups can be used with bootstrap Badages.
- https://www.jquery-az.com/bootstrap4/demo.php?ex=78.0_8

First item	Second item	Third item	Fourth item
------------	-------------	------------	-------------

JavaScript
Bootstrap
HTML
CSS
Angular JS

Bootstrap Dropdowns

- A dropdown menu is a toggleable menu that allows the user to choose one value from a predefined list.
- 1. Add **.dropdown class** indicates a dropdown menu.
- 2. To open the dropdown menu, use a button or a link with a class of **.dropdown-toggle** and the **data-toggle="dropdown"** attribute.
- 3. Add the **.dropdown-menu** class to a `<div>` element to actually build the dropdown menu. Then add the **.dropdown-item** class to each element (links or buttons) inside the dropdown menu.

```
<div class="dropdown">
```

```
<a href="#" class="dropdown-toggle" data-toggle="dropdown">Dropdown</a>
```

```
<div class="dropdown-menu">
```

```
  <a href="#" class="dropdown-item">Action</a>
```

```
  <a href="#" class="dropdown-item">Another action</a>
```

```
</div>
```

```
</div>
```

Dropdown ▼

Action

Another action

- **.dropdown-item active**: Highlight a specific dropdown item with the **.active** class
- **.dropdown-item disabled**: disable an item in the dropdown menu
- **.dropdown-header**: used to add headers inside the dropdown menu
- **.dropdown-divider**: separate links inside the dropdown menu with a thin horizontal border

Bootstrap Inputs

- Bootstrap supports all the HTML5 input types:
 - text, password, datetime, datetime-local, date, month, time, week, number, email, url, search, tel, and color.

-

```
<div class="form-group">  
  <label for="usr">Name:</label>  
  <input type="text" class="form-control" id="usr">  
</div>  
<div class="form-group">  
  <label for="pwd">Password:</label>  
  <input type="password" class="form-control" id="pwd">  
</div>
```

Bootstrap Forms

Email

Password

☐ Remember me

Sign in

- Every `<input>` should be given `.form-control`

```
<form>
  <div class="form-group">
    <label class="control-label"></label>
    <input class="form-control">
  </div>
</form>
```

```
<form>
<div class="form-group">
  <label for="inputEmail">Email</label>
<input type="email" class="form-control"
id="inputEmail" placeholder="Email"> </div>
  <div class="form-group">
<label for="inputPassword">Password</label>
<input type="password" class="form-control"
id="inputPassword" placeholder="Password">
</div>
  <div class="form-group">
<label class="form-check-label">
<input type="checkbox"> Remember me</label>
</div>
<button type="submit" class="btn btn-
primary">Sign in</button>
</form>
```

Also note that we add a `.form-label` class to each label element to ensure correct padding.

Bootstrap Buttons

- Buttons are used to trigger some action when pressed; almost every webpage will use buttons at some point.
- Button Styles
 - Bootstrap provides seven styles of buttons with the following classes:
 - .btn-default
 - .btn-primary
 - .btn-success
 - .btn-info
 - .btn-warning
 - .btn-danger
 - .btn-link

Bootstrap Button -Example



And the source code to generate the outputs looks like this:

```
1 <button type="button" class="btn btn-primary">Primary</button>
2 <button type="button" class="btn btn-secondary">Secondary</button>
3 <button type="button" class="btn btn-success">Success</button>
4 <button type="button" class="btn btn-danger">Danger</button>
5 <button type="button" class="btn btn-warning">Warning</button>
6 <button type="button" class="btn btn-info">Info</button>
7 <button type="button" class="btn btn-light">Light</button>
8 <button type="button" class="btn btn-dark">Dark</button>
9
10 <button type="button" class="btn btn-link">Link</button>
```

Button Sizes

- Bootstrap provides four button sizes with the following classes:

.btn-lg

.btn-md

.btn-sm

.btn-xs

Active/Disabled Buttons

A button can be set to an active (appear pressed) or a disabled (unclickable) state:

- The class `.active` makes a button appear pressed, and the class `.disabled` makes a button unclickable:
- add the class `.active` on input's `<label>` if we want options pre-checked by default

Example

```
<label class="btn btn-primary active">
```

Toggle Button

- add the attribute `data-toggle="buttons"`

Single Toggle button:

```
<button type="button" class="btn btn-primary" data-toggle="button"
autocomplete="off">Single Toggle Button</button>
"btn-group-toggle"
```



Single Toggle Button

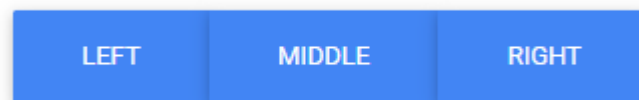
Button Groups

- Bootstrap allows to group a series of buttons together (on a single line) in a button group

Use a `<div>` element with class `.btn-group` to create a button group

`class="btn-group"`

- `<div class="btn-group">`
 `<button type="button" class="btn btn-primary">Left</button>`
 `<button type="button" class="btn btn-primary">Middle</button>`
 `<button type="button" class="btn btn-primary">Right</button>`
 `</div>`

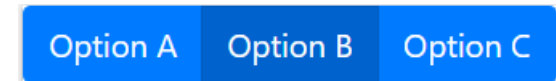


Radio Button Grouping

- Add the attribute `data-toggle="buttons"` to a **group of Radio buttons** for checkbox style toggling on button groups.

- Example

- ```
<div class="btn-group btn-group-toggle" data-toggle="buttons">
 <label class="btn btn-primary">
 <input type="radio" name="options" autocomplete="off"> Option A
 </label>
 <label class="btn btn-primary active">
 <input type="radio" name="options" autocomplete="off" checked> Option B
 </label>
 <label class="btn btn-primary">
 <input type="radio" name="options" autocomplete="off"> Option C
 </label>
</div>
```



<https://www.tutorialrepublic.com/codelab.php?topic=bootstrap&file=buttons-radio>

# Button Group- Sizing

Add btn-group-\* to each btn-group instead of applying button sizing classes to every button in a group. Following btn-group-\* sizes are available

- btn-group-lg
- btn-group-sm
- btn-group-vertical
- btn-group btn-group-justified
- Example

---

Left Middle Right

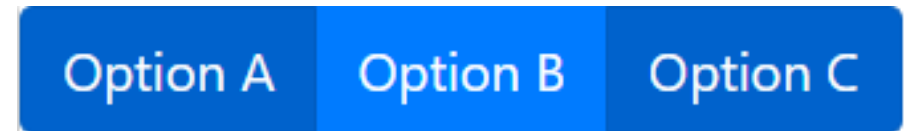
```
<div class="btn-group btn-group-sm" role="group">
 <button type="button" class="btn btn-unique btn-sm">Left</button>
 <button type="button" class="btn btn-unique btn-sm">Middle</button>
 <button type="button" class="btn btn-unique btn-sm">Right</button>
</div>
```

# CheckBox Grouping

Add the attribute `data-toggle="buttons"` to a **group of checkboxes** for checkbox style toggling on button groups.

- Example

- ```
<div class="btn-group btn-group-toggle" data-toggle="buttons">  
  <label class="btn btn-primary active">  
    <input type="checkbox" name="options" autocomplete="off" checked> Option A  
  </label>  
  <label class="btn btn-primary">  
    <input type="checkbox" name="options" autocomplete="off"> Option B  
  </label>  
  <label class="btn btn-primary active">  
    <input type="checkbox" name="options" autocomplete="off" checked> Option C  
  </label>  
</div>
```



<https://www.tutorialrepublic.com/codelab.php?topic=bootstrap&file=buttons-checkbox>

Bootstrap Nav

- Use **.nav class** to create a basic navigation menu, like this
- **class=nav-item** will add items to the navigations.

```
<nav class="nav">
```

```
<a href="#" class="nav-item nav-link active">Home</a>
```

```
<a href="#" class="nav-item nav-link">Profile</a>
```

```
<a href="#" class="nav-item nav-link">Messages</a>
```

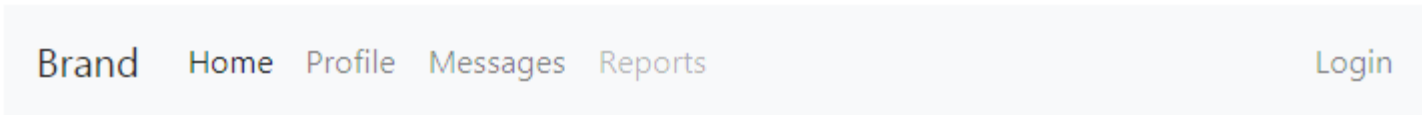
```
<a href="#" class="nav-item nav-link disabled" tabindex="-1">Reports</a>
```

```
</nav>
```

Home Profile Messages Reports

Bootstrap Navigation Bar

- A navigation bar is a navigation header that is placed at the top of the page.
- Use **.navbar** class to make a Navigation bar:



- **.navbar-expand-xl | lg | md | sm** (stacks the navbar vertically on extra large, large, medium or small screens).
- To add links inside the navbar,
- use a `<ul>` element with `class="navbar-nav"`.
- Then add `<li>` elements with a `.nav-item` class followed by an `<a>` element with a `.nav-link` class

Bootstrap Tabs

- Tab based navigations displays huge amount of content within a small area
- separates content into different panes where each pane is viewable one at a time.
- The user can quickly access the content through switching between the panes without leaving the page



```
<ul class="nav nav-tabs">
  <li class="nav-item">
    <a href="#" class="nav-link active">Home</a></li>
    <li class="nav-item">
      <a href="#" class="nav-link">Profile</a>
    </li>
    <li class="nav-item">
      <a href="#" class="nav-link">Messages</a>
    </li>
</ul>
```

Bootstrap Pagination

- Pagination is the process of organizing content by dividing it into separate pages.
- If you have a web site with lots of pages, you may wish to add some sort of pagination to each page.
- To create a basic pagination
 1. add the .pagination class to an element.
 2. add the .page-item to each element
 3. a .page-link class to each link inside

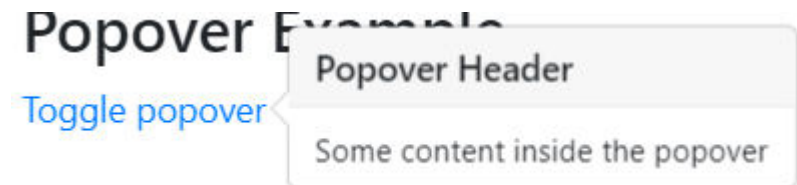
The **.active** class is used to "highlight" the current page

<https://www.tutorialrepublic.com/codelab.php?topic=bootstrap&file=pagination>

Bootstrap-Popovers

- The Popover component is similar to tooltips; it is a **pop-up box** that appears when the **user clicks on an element**. The difference is that the popover can contain much more content.
- To create a popover, add the **data-toggle="popover"** attribute to an element.
- Use the **title attribute** to specify the header text of the popover,
- Use the **data-content** attribute to **specify the text** that should be displayed inside the popover's body.
- Popovers must be initialized with jQuery: select the specified element and call the `popover()` method.
- `<a href="#" data-toggle="popover" title="Popover Header" data-content="Some content inside the popover">Toggle popover</a>`
- https://www.w3schools.com/bootstrap4/tryit.asp?filename=trybs_popover&stacked=h

```
<script>
$(document).ready(function(){
    $('[data-toggle="popover"]').popover();
});
</script>
```



Bootstrap-Alerts

Alerts are used to display some sort of message such as a warning or a notification. Here are a few examples of Bootstrap alerts:

```
<div class="alert alert-primary" role="alert">  
  A simple primary alert—check it out!  
</div>  
<div class="alert alert-secondary" role="alert">  
  A simple secondary alert—check it out!  
</div>  
<div class="alert alert-success" role="alert">  
  A simple success alert—check it out!  
</div>
```

A simple primary alert—check it out!

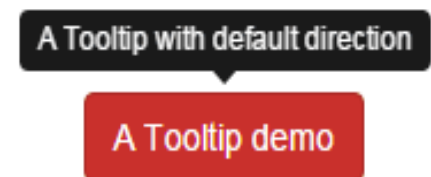
A simple secondary alert—check it out!

A simple success alert—check it out!

Bootstrap-Tooltip

- The Tooltip component **is small pop-up box** that appears when the user *moves the mouse pointer over an element*.
- To create a tooltip, add the **data-toggle="tooltip"** attribute to an element.
- Use the **title attribute** to specify the text that should be displayed inside.
- `<a href="#" data-toggle="tooltip" title="Hooray!">Hover over me</a>`
- Tooltips must be initialized with jQuery: select the specified element and call the `tooltip()` method.
- https://www.w3schools.com/bootstrap4/tryit.asp?filename=trybs_tooltip&stacked=h

```
<script>
$(document).ready(function(){
    $('[data-toggle="tooltip"]').tooltip();
});
</script>
```



Bootstrap-Toasts

- The toast component is **like an alert box** that is only shown for a couple of seconds
- when something happens (i.e. when the user clicks on a button, submits a form, etc.).

Bootstrap Classes

To create a toast, use the **.toast class**, and add a **.toast-header** and a **.toast-body** inside of it:

- Toasts must be initialized with jQuery: select the specified element and call the toast() method inside <script> tag.

```
<script>
```

```
$(document).ready(function(){
```

```
  $("#myBtn").click(function(){
```

```
    $('.toast').toast('show');
```

```
  });
```

```
});
```

```
</script>
```

Toast Header

5 mins ago X

Some text inside the toast body

Bootstrap Toast-Example

- `<button type="button" class="btn btn-primary" id="myBtn">Show Toast</button>`

`<div class="toast">`

`<div class="toast-header">`

Toast Header

`</div>`

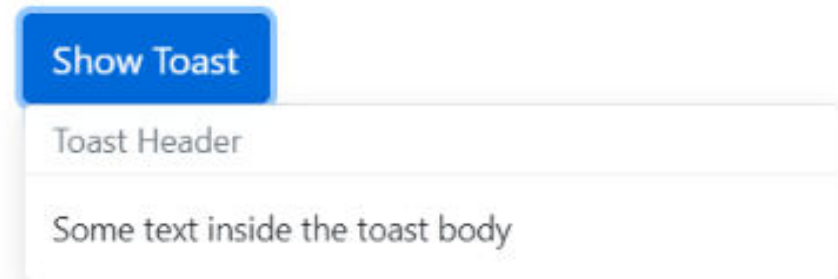
`<div class="toast-body">`

Some text inside the toast body

`</div>`

`</div>`

- https://www.w3schools.com/bootstrap4/tryit.asp?filename=trybs_toast&stacked=h



Bootstrap Collapse

- Collapsibles are used to **hide and show large amount of content**.
- The **.collapse class** indicates a collapsible element (this is the content that will be shown or hidden with a click of a button).
- To control (show/hide) the collapsible content,
 1. add the data-toggle="collapse" attribute to an <a> or a <button> element.
 2. Then add the data-target="#id" attribute to connect the button with the collapsible content (<div id="demo">).

```
<button data-toggle="collapse" datatarget="#demo">Collapsible</button>
```

```
<div id="demo" class="collapse">
```

```
  Lorem ipsum dolor text....
```

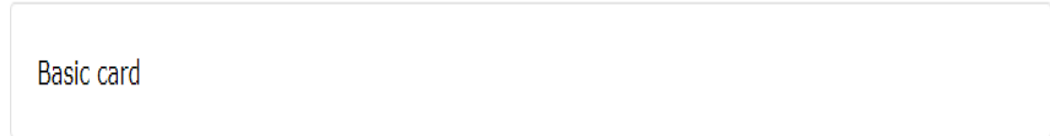
```
</div>
```

By default, the collapsible content is hidden, .show class to show the content by default



Bootstrap Cards

- A card in Bootstrap 4 is a **bordered box with some padding around its content**. It includes options for headers, footers, content, colors, etc.
- Use **.card class**, and **.card-body class** for content of the cards.
 - `<div class="card">`
 `<div class="card-body">Basic card</div>`
 `</div>` inside
- The **.card-header** class adds a heading to the card
- The **.card-footer** class adds a footer to the card
- The **.card-title** adds card titles to any heading element.
- The **.card-text** class is used to remove bottom margins for a `<p>` element if it is the last child (or the only one) inside **.card-body**.
- The **.card-link** class adds a blue color to any link, and a hover effect.
- Add **.card-img-top** or **.card-img-bottom** to an `<img>` to place the image at the top or at the bottom inside the card.
- The **.card-columns** class creates a masonry-like grid of cards (like pinterest).
- The **.card-deck** class creates a grid of cards that are of equal height and width. The layout will automatically adjust as you insert more cards.



Bootstrap Cards examples

Header and Footer

Header

Content

Footer

Card Columns

Some text inside the first card

Some text inside the second card

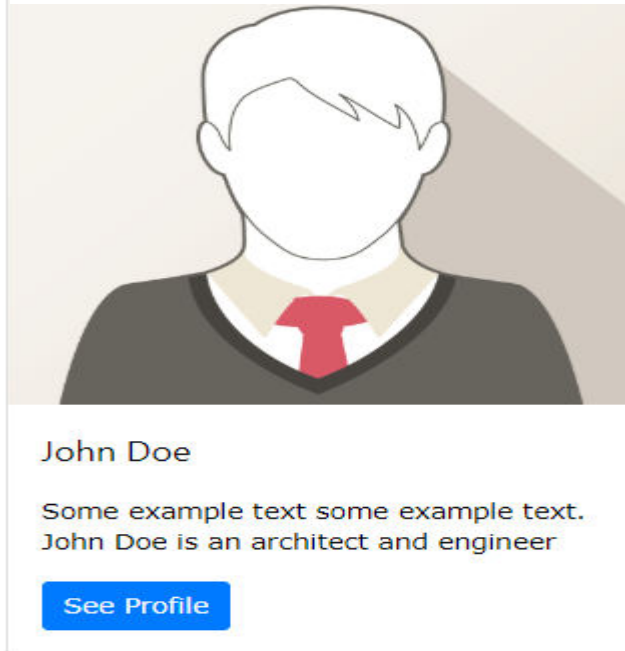
Card Deck

Some text inside the first card

Some more text to increase the height

Some more text to increase the height

Some more text to increase the height



Contextual Cards

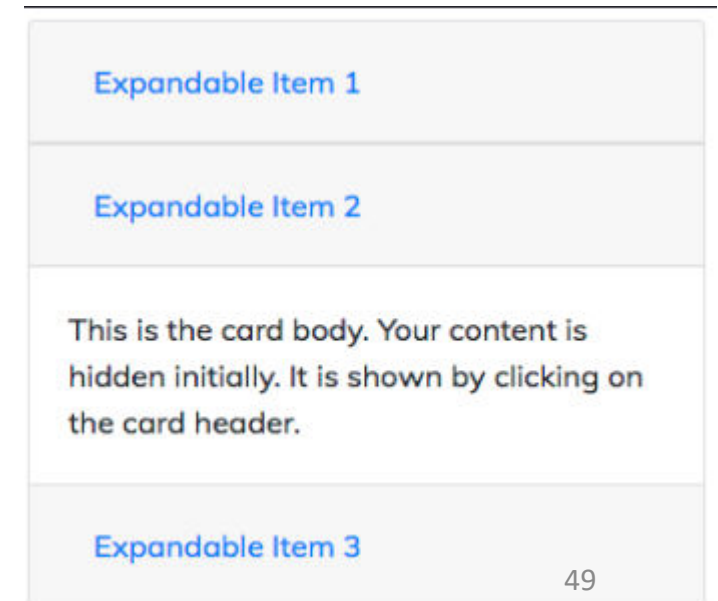
To add a background color the card, use contextual classes

.bg-primary, .bg-success, .bg-info, .bg-warning, .bg-danger, .bg-secondary, .bg-dark and .bg-light.

```
<div class="card border-dark mb-3" style="width: 20rem;">
  
  <div class="card-body text-dark">
    <h5 class="card-title">Dark card title</h5>
    <p class="card-text">Lorem ipsum dolor sit
    amet, consectetur adipiscing elit.</p>
  </div>
</div>
```


Bootstrap Accordion

- Accordions are useful when you want to toggle between hiding and showing large amount of content.
- Can make a Bootstrap panel collapse and reappear
- Collapsible are useful for **hiding/showing** large amounts of content.
- Bootstrap **accordion** is a component that holds multiple pieces of collapsible data and only displays **one** at a time.
- Use **class="accordion"**
- **Use Accordion with bootstrap card/panel**
- A Bootstrap accordion has multiple collapsible cards, and only one of them can be shown at a time.



Bootstrap Accordion Example

```
<div class="accordion" id="accordionExample">
```

```
  <div class="card">
```

```
    <div class="card-header" id="headingOne">
```

```
      <h2 class="mb-0">
```

```
        <button type="button" class="btn btn-link" data-toggle="collapse" data-target="#collapseOne">1. What is HTML?</button> </h2></div>
```

```
    <div id="collapseOne" class="collapse" aria-labelledby="headingOne" data-parent="#accordionExample">
```

```
      <div class="card-body">
```

```
        <p>HTML stands for HyperText Markup Language. HTML is the standard markup language for describing the structure of web pages. <a href="https://www.tutorialrepublic.com/html-tutorial/" target="_blank">Learn more.</a></p></div></div></div>
```

```
  <div class="card">
```

```
    <div class="card-header" id="headingtwo">
```

```
      <h2 class="mb-0">
```

```
        <button type="button" class="btn btn-link" data-toggle="collapse" data-target="#collapseOne">1. What is HTML?</button> </h2></div>
```

```
    <div id="collapseOne" class="collapse" aria-labelledby="headingOne" data-parent="#accordionExample">
```

```
      <div class="card-body">
```

```
        <p>Bootstrap is a sleek, intuitive, and powerful front-end framework for faster and easier web development. It is a collection of CSS and HTML conventions. <a href="https://www.tutorialrepublic.com/twitter-bootstrap-tutorial/" target="_blank">Learn more.</a></p></div></div></div>
```

[1. What is HTML?](#)

HTML stands for HyperText Markup Language. HTML is the standard markup language for describing the structure of web pages. [Learn more.](#)

[2. What is Bootstrap?](#)

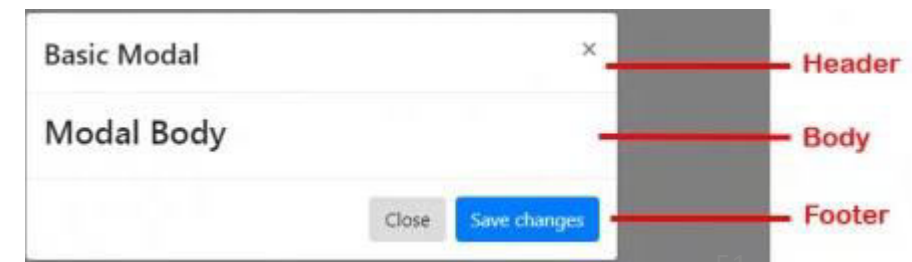
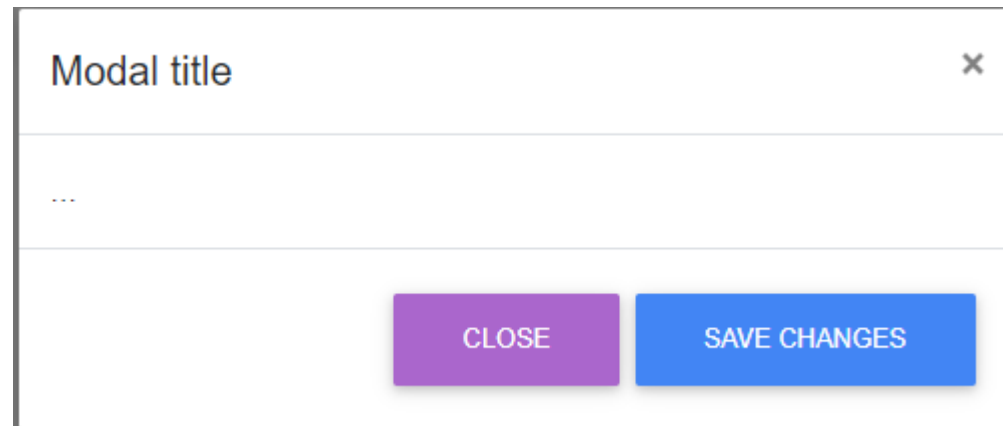
[3. What is CSS?](#)

Bootstrap Modals

- easiest and handiest ways to build **dialog windows**.
- **Lightweight, *multi-purpose JavaScript popup* that's customizable and responsive.**
- **They can be used to display alert popups, videos, and images in a website.**
- Bootstrap Modals are divided into three primary sections: the header, body, and footer .
- On click, Modal will be shown on the screen.



LAUNCH DEMO MODAL



Modal-Overview

- **The "Trigger" part:**

- To trigger the modal window, We need to use a button or a link.
- Then include the two data-* attributes with button tags:
- **data-toggle="modal"** opens the modal window
- **data-target="#myModal"** points to the id of the modal

- **The “Modal” part:**

- Need a parent <div> of modal with an ID that is the same as the value of the data-target attribute used to trigger the modal ("myModal").
- **Modal Classes**
- "modal-content-
- .modal-title
- .modal-body
- .modal-footer

Modal-Example

```
<!-- Trigger the modal with a button -->
```

```
<button type="button" class="btn btn-info btn-lg" data-toggle="modal" data-target="#myModal">Show </button>
```

```
<!-- Modal -->
```

```
<div id="myModal" class="modal fade" role="dialog">
```

```
  <div class="modal-dialog">
```

```
    <!-- Modal content-->
```

```
    <div class="modal-content">
```

```
      <div class="modal-header">
```

```
        <button type="button" class="close" data-dismiss="modal">&times;</button>
```

```
        <h4 class="modal-title">Modal Header</h4>
```

```
      </div>
```

```
      <div class="modal-body">
```

```
        <p>Some text in the modal.</p>
```

```
      </div>
```

```
      <div class="modal-footer">
```

```
        <button type="button" class="btn btn-default" data-dismiss="modal">Close</button>
```

```
      </div>
```

```
    </div>
```

```
  </div>
```

```
</div>
```

Bootstrap – Progress Bar

- Progress bars can be used for **showing the progress of a task/action** to the users.
- Use a `<div>` with `class= “progress”`
- Use a child `<div>` with `class = “progress-bar”`
- Example

```
<div class="progress">
```

```
<div style="width: 50%;" class="progress-bar">60%</div>
```

```
</div>
```



Bootstrap – Progress Bar

- To create the striped progress bar just add an extra class `.progress-bar-striped` to the `.progress-bar` element,



- place multiple bars into the same progress bar to stack them.

```
<div class="progress">  
  <div class="progress-bar bg-success" style="width: 40%">  
    Program Files (40%)  
  </div>  
  <div class="progress-bar bg-warning" style="width: 25%">  
    Residual Files (25%)  
  </div>  
  <div class="progress-bar bg-danger" style="width: 15%">  
    Junk Files (15%)  
  </div>  
</div>
```



Changing Progress Bar Value Dynamically

- Need javascript or JQuery
- Need to update width of progress bar in intervals
- Write a javascript/jquery function to do so.
- Demo
 - <https://www.tutorialrepublic.com/codelab.php?topic=bootstrap&file=dynamically-change-bootstrap-progress-bar-value>
 - <https://www.geeksforgeeks.org/creating-progress-bar-using-javascript/>

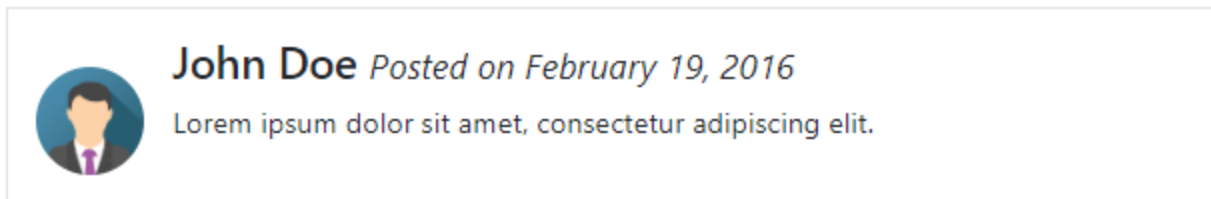
Bootstrap Spinners

- Spinners are typically loading icons & used to show the loading state.
- To create a spinner/loader, use the **.spinner-border** class:
- Use any **text color utilites** to add a color to the spinner using these classes such as **.text-muted**, **.text-primary**, **.text-secondary**, **.text-success**, **.text-info** etc.
- ```
<div class="spinner-border text-primary"></div>
<div class="spinner-border text-success"></div>
<div class="spinner-border text-info"></div>
```
- <https://www.tutorialrepublic.com/codelab.php?topic=bootstrap&file=spinners-inside-buttons>



# Bootstrap Media Objects

- align **media objects (like images or videos)** together with content.
- Media objects are often used to **display blog comments, tweets** and so on.
- To create a media object,
  1. add the `.media` class to a container element,
  2. place media content inside a child container with the `.media-body` class.
- ```
<div class="media border p-3">  
    
  <div class="media-body">  
    <h4>John Doe <small><i>Posted on February 19, 2016</i></small></h4>  
    <p>Lorem ipsum...</p>  
  </div>  
</div>
```



Bootstrap Carousel

- The Carousel is a slideshow for cycling through elements.

Class	Description
<code>.carousel</code>	Creates a carousel
<code>.carousel-indicators</code>	Adds indicators for the carousel. These are the little dots at the bottom of each slide (which indicates how many slides there are in the carousel, and which slide the user are currently viewing)
<code>.carousel-inner</code>	Adds slides to the carousel
<code>.carousel-item</code>	Specifies the content of each slide
<code>.carousel-control-prev</code>	Adds a left (previous) button to the carousel, which allows the user to go back between the slides
<code>.carousel-control-next</code>	Adds a right (next) button to the carousel, which allows the user to go forward between the slides
<code>.carousel-control-prev-icon</code>	Used together with <code>.carousel-control-prev</code> to create a "previous" button
<code>.carousel-control-next-icon</code>	Used together with <code>.carousel-control-next</code> to create a "next" button
<code>.slide</code>	Adds a CSS transition and animation effect when sliding from one item to the next. Remove this class if you do not want this effect

Add elements inside `<div class="carousel-caption">` within each `<div class="carousel-item">` to create a caption for each slide

Bootstrap Carousel Example

```
<div id="demo" class="carousel slide" data-ride="carousel">
```

```
  <ul class="carousel-indicators">
```

```
    <li data-target="#demo" data-slide-to="0" class="active"></li>
```

```
    <li data-target="#demo" data-slide-to="1"></li>
```

```
    <li data-target="#demo" data-slide-to="2"></li> </ul>
```

```
<div class="carousel-inner">
```

```
<div class="carousel-item active">
```

```
  <div class="carousel-caption"> <h3>Los Angeles</h3> <p>We had such a great time in LA!</p> </div> </div>
```

```
div class="carousel-item ">
```

```
  <div class="carousel-caption"> <h3>Chicago</h3> <p>Thank you, Chicago!</p> </div> </div>
```

```
<a class="carousel-control-prev" href="#demo" data-slide="prev">
```

```
  <span class="carousel-control-prev-icon"></span>
```

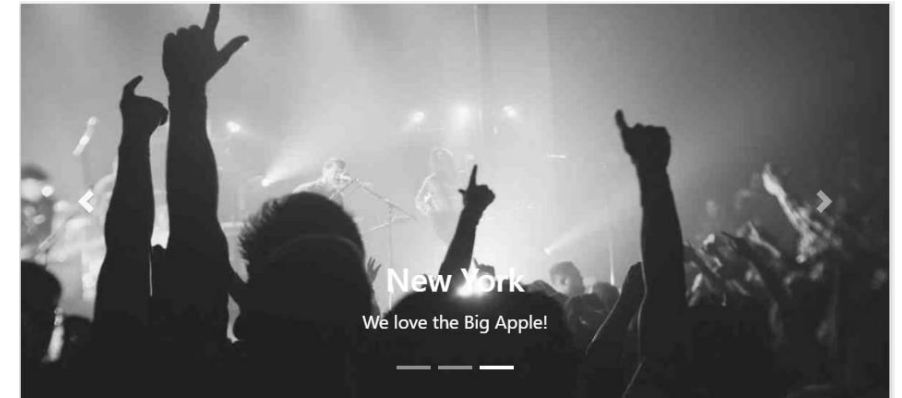
```
</a>
```

```
<a class="carousel-control-next" href="#demo" data-slide="next">
```

```
  <span class="carousel-control-next-icon"></span>
```

```
</a>
```

```
</div>
```



References

- [W3schools.com](https://www.w3schools.com)
- <https://www.tutorialrepublic.com/twitter-bootstrap-tutorial/>